Comparing Shared Representations and Explicit Communication in LLM Agents

Bryan Nsoh, Gagandeep Kaur, Joseph Gyasi Bronyah, Sobhan Bahrami

Abstract

We investigate two paradigms of multi-agent coordination using Large Language Models (LLMs): explicit communication via message passing and implicit coordination via shared representation. We populate a bandwidth-constrained, partially observable grid world ($R_{vis} = 1$) with GPT-5-Mini agents tasked with navigating a linear bottleneck. We find that shared representation achieves perfect search performance, with global map sharing reaching 100% agent completion compared to 84% in the silent baseline ($p < 0.001$). However, this perfect environmental knowledge does not resolve traffic conflicts; agents with global maps collide just as frequently (7.1 per run) as blind baseline agents (6.6 per run). Conversely, explicit communication strategies, whether structured or freeform, fail to provide a statistically significant improvement over their baseline in success rate or collision avoidance ($p > 0.5$). We conclude that while shared state solves search, distinct conflict-resolution protocols are required to manage the traffic bottlenecks it creates.

1 Introduction

Large language models are increasingly deployed as collections of interacting agents rather than as isolated systems. Frameworks such as AutoGen and MetaGPT make it easy to instantiate role-specialized LLM agents that solve tasks by exchanging messages, while more recent systems like MegaAgent focus on scaling these patterns to hundreds of agents with hierarchical task decomposition and shared message pools [10, 3, 8]. In these settings, "communication" is usually understood in the intuitive sense: agents send natural language or lightly structured messages to one another, and coordination emerges from the conversational protocol plus the underlying model's priors about teamwork.

In parallel, there is growing interest in using LLM agents as inhabitants of synthetic worlds. Social simulation work populates toy towns or social networks with dozens or thousands of LLM-driven characters and studies their interactions over time [6], while other projects treat LLMs as embodied controllers in games or virtual environments. The Abstraction and Reasoning Corpus (ARC-AGI) offers grid-based visual puzzles that probe pattern discovery and generalization, becoming a touchstone for evaluating reasoning in models that operate on JSON or textual encodings of small grids [2, 1]. Similarly, embodied agents like Voyager in Minecraft showcase how a single LLM can drive long-horizon behavior in an environment surfaced entirely as text or abstract state [9, 7].

A third strand of research studies communication and coordination under stronger experimental control, mostly in the multi-agent reinforcement learning community. Classic work on differentiable communication architectures used small partially observable grid worlds to show how agents can learn when and what to broadcast. More recent approaches add language grounding and human interpretability, for example by training policies to imitate messages produced by LLM teammates [4]. These studies give careful control over reward and topology but typically operate with learned, opaque symbol channels rather than the natural language priors of off-the-shelf models.

At the systems level, there is growing interest in "communication" as sharing internal representations rather than messages. KV cache reuse and prefix caching can dramatically reduce prefill latency, and recent work extends this to multiple fine-tuned models by selectively recomputing only specific layers [5]. In these systems, no natural language flows between agents; they coordinate indirectly by reusing a shared latent state. This notion of communication via shared representations is conceptually quite different from conversational protocols, yet it is arguably closer to how large model deployments scale in practice.

However, a standardized benchmark for measuring inter-agent communication efficiency remains elusive. Complex environments like Minecraft or multi-agent games introduce confounding variables—physics engines, crafting mechanics, open-ended objectives—that obscure whether failures stem from poor communication protocols or poor strategic planning. Conversely, purely conversational benchmarks lack the spatial constraints (collisions, bottlenecks, resource contention) that make coordination necessary in embodied settings. There is a gap for an environment that is abstract enough to isolate communication variables, yet scalable enough to test protocols ranging from simple pairs to large swarms.

We address this gap by constructing `LLMGrid`, a modular grid world designed specifically to stress-test agent coordination under controlled conditions. The environment is rendered as an ASCII map and accompanying JSON, populated by identical LLM agents. Each agent perceives only its immediate surroundings ($R_{vis} = 1$). The action space is minimal: move cardinal directions, stay, or use a communication channel. The environment's modularity—tunable grid size, obstacle density, agent count, visibility radius, and communication bandwidth—allows researchers to isolate individual variables (e.g., message latency, radio range) while holding others constant.

Within this environment, we instantiate two coordination paradigms to demonstrate its diagnostic utility. In the first, communication is explicit message passing: agents send structured JSON packets or freeform natural language messages to teammates, delivered through the textual interface. In the second, communication is implicit via shared representations: agents maintain internal maps that are either merged when within radio range or synchronized globally. This setup allows us to ask: when given explicit channels, do agents discover useful protocols? When given only a shared representation, do they exploit it? And how do these modes compare to having no communication at all?

2 The LLMGrid Environment

We utilize `LLMGrid`, a deterministic, discrete-time simulation engine. The environment topology is a 30×10 grid world generated via a recursive backtracking algorithm. This specific layout, the "Long Corridor," was selected for its high connectivity index combined with narrow chokepoints (1-2 cells wide) that mandate agent interaction to prevent blocking.

2.1 Simulation Dynamics

Five agents operate within the grid with a fixed turn budget ($T = 100$). The simulation proceeds in synchronized turns where the engine solicits actions from all agents, resolves conflicts, and advances the state.

- **Visibility ($R_{vis} = 1$):** Agents operate under severe partial observability, perceiving only their own coordinate and the four immediate cardinal neighbors.
- **Movement & Conflict:** Agents plan moves (N, E, S, W) or $STAY$. If multiple agents attempt to enter the same cell, or if two agents attempt to swap positions, the engine resolves this as a `BLOCK_AGENT` outcome. No displacement occurs, and the turn is consumed.
- **Latency:** To model realistic processing and network constraints, explicit messages sent at turn t are delivered to recipients at turn $t + 1$.

2.2 The Agent Interface

Agents interact with the environment via a structured JSON prompt-response loop. They do not receive visual input; instead, they receive a local sensor patch and a persistent memory object (an ASCII map). Figure 1 illustrates the input-output contract for the Communication conditions.

The two experimental branches differ in navigation affordances, not merely communication channels. In the Communication branch, agents receive a `goal_sensor` that provides noisy bearing information (octants: N, NE, E, etc.) and distance buckets (FAR, MID, NEAR). This allows agents to navigate toward the goal even before discovering it visually. The agent's output includes an explicit `message` payload that is broadcast to teammates within radio range.

In the Shared Representation branch, the `goal_sensor` is removed entirely. Agents must explore the grid blindly until they visually discover the goal symbol. The message channel is also disabled; instead, the internal map state is automatically synchronized with other agents (either upon contact in the Radio condition, or globally in the Global condition). This design tests whether shared environmental knowledge can substitute for both goal guidance and explicit messaging. Figure 2 shows the environment during a typical run, while Figure 3 illustrates the fog-of-war partial observability.

3 Experimental Design

We executed 90 total episodes ($N = 15$ per condition). The variability in the experiment is controlled by the random placement of agents and the sampling noise of the model.

- **Map Topology:** The wall layout is fixed for all runs to ensure consistent pathing difficulty.
- **Agent Spawns (Seeds 13–17):** Five distinct integers used to initialize the pseudo-random number generator that places the 5 agents at start positions. These positions are randomized but guaranteed to be valid (non-wall) and equidistant from the goal relative to the grid difficulty.
- **Replicas ($N = 3$):** For each Run Seed and Condition, we execute 3 independent runs. Since the engine is deterministic, variance between replicas is driven entirely by the probabilistic sampling of the LLM (Temperature 1.0).

We group the six conditions into two investigation branches, summarized in Table 1. Each branch utilizes a specific control baseline matching its interface (Sensor vs. ASCII Map) to ensure fair comparison.

4 Results

We present statistical outcomes for both branches using the Mann-Whitney U test, comparing each experimental

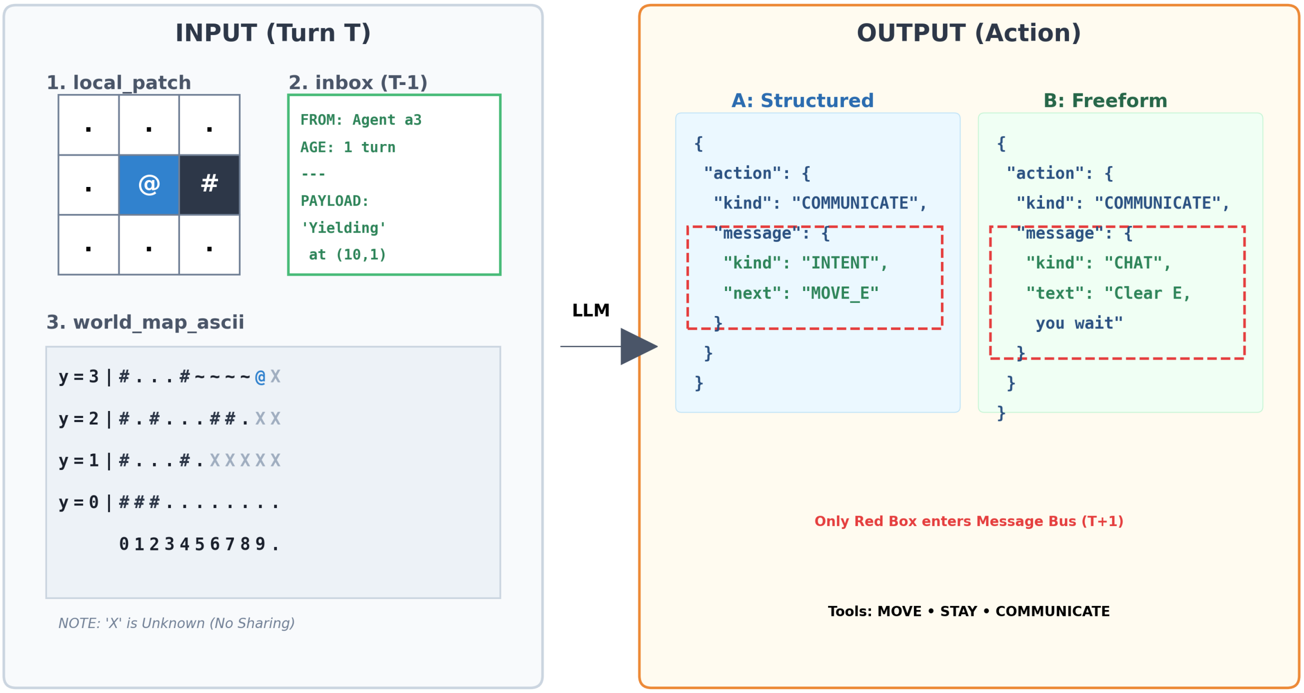

Figure 1: The Coordination Interface. Left: The agent receives a sparse observation containing a 3×3 local patch ($R = 1$), an inbox containing messages sent by peers in the previous turn ($T-1$), and a world map showing discovered tiles. Right: The output schema varies by condition. Structured agents must select from a rigid set of intents or requests, while Freeform agents generate natural language strings. Only the message payload (red box) enters the message bus at turn $T + 1$.

condition against its respective baseline. The choice between one-tailed and two-tailed tests was driven by the hypothesis directionality. For Map Sharing (search problem), we used one-tailed tests (alternative='greater') because the hypothesis was specifically directional: does shared knowledge improve search? We were testing if Global/Radio was better than Baseline. For Communication (coordination problem), we used two-tailed tests (alternative='two-sided') because the effect was uncertain. Explicit communication could plausibly help (by coordinating) or hurt (by wasting turns on message latency). We needed to detect significant deviation in either direction. Table 2 provides a consolidated view of all statistical tests performed.

4.1 Shared Representation Achieves Perfect Search

In the Map Sharing branch, agents must explore blindly without a goal sensor. The silent Baseline B achieved an agent completion rate of 84.0% (63/75 agents). This relatively high baseline suggests that random walk is a viable strategy in this topology given 100 turns. However, **Global Map Sharing** eliminates the remaining 16% of failures, achieving 100% completion (75/75 agents). This improvement is statistically significant ($U = 180.0, p < 0.001$).

Radio Sync provided a marginal improvement (88.0% completion), but the benefit did not reach statistical significance ($p = 0.18$). As illustrated in Figure 4, this is due to a "Knowledge Plateau": knowledge sharing creates an initial burst of discovery, but as agents disperse beyond the radio range ($r = 2$), the collective intelligence fractures into isolated exploration.

4.2 Explicit Communication Fails to Coordinate

In the Communication branch, agents possess a goal sensor, resulting in a baseline completion rate of 57.3%. The lower baseline compared to Branch 2 is attributed to the goal sensor encouraging agents to crowd into optimal paths, creating contention that blind agents avoid by wandering.

Explicit communication failed to improve this outcome. **Freeform** communication achieved 62.7% success and **Structured** achieved 56.0% success; neither result is statistically distinguishable from the baseline ($p > 0.5$). Furthermore, messaging failed to reduce physical conflicts. The Freeform condition averaged 18.9 collisions per run,

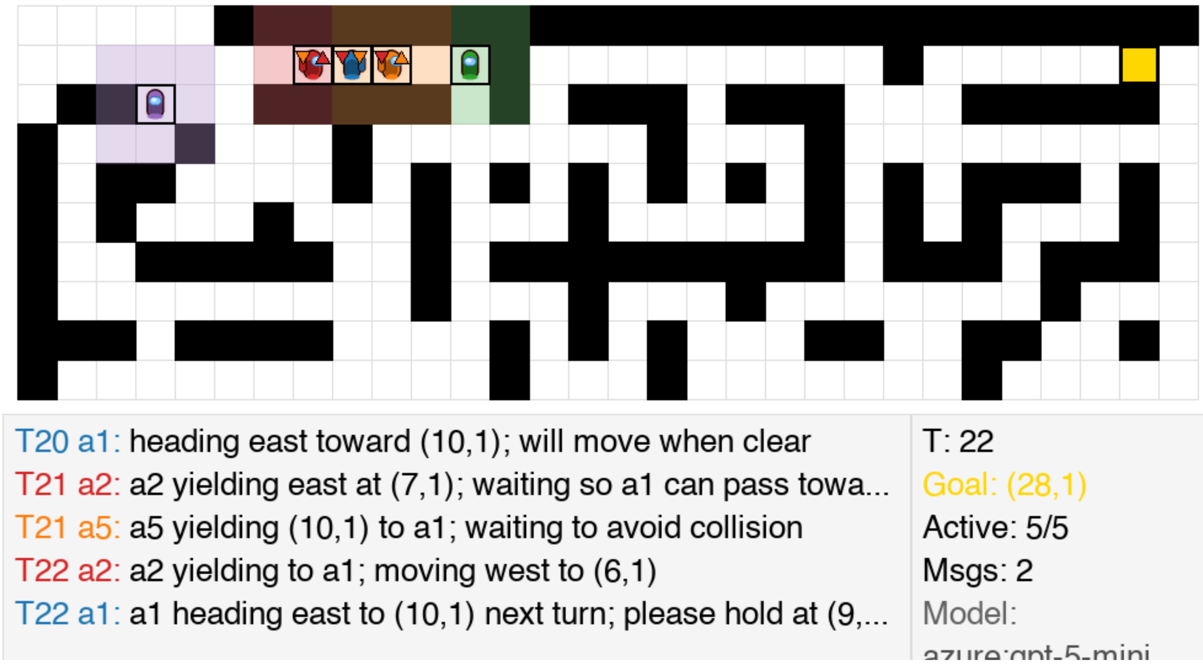

Figure 2: The "Long Corridor" Environment in Action. Snapshot from a Communication baseline run (Freeform condition) showing 5 agents navigating the maze. The Among Us sprites with directional indicators show agent positions and orientations. Visibility auras (colored overlays) indicate each agent's limited $R_{vis} = 1$ perception range. The goal gradient (green intensity) shows distance to the target. Note: This visualization is from the Communication branch; the Map-Sharing branch uses identical maze topology but different agent interfaces.

Table 1: Unified Experimental Design Matrix ($N = 15$ runs per condition)

Paradigm	**Condition**	**Map**	**Comm.**	**Protocol**	**Hypothesis**
Branch 1 (Explicit)	Baseline A	Local	None	Silent control (Sensor)	Control for Comms
	Structured	Local	$r = 2$	Schema: INTENT/REQUEST	Strict schemas reduce ambiguity
	Freeform	Local	$r = 2$	Natural Language (<96 chars)	Language handles edge cases
Branch 2 (Implicit)	Baseline B	Local	None	Silent control (ASCII Map)	Control for Search
	Radio Sync	Radio	None	Maps merge on contact	Local sharing aids discovery
	Global	Infinite	None	Oracle map shared instantly	Perfect info maximizes efficiency

Table 2: Statistical Summary (Mann-Whitney U)

Comparison	**Metric**	**p**	**Sig.**
Global vs. Base B	Finished	<.001	***
Radio vs. Base B	Finished	.18	n.s.
Freeform vs. Base A	Success	.58	n.s.
Structured vs. Base A	Success	.93	n.s.
Freeform vs. Base A	Collisions	.95	n.s.
Global vs. Base B	Collisions	.83	n.s.

compared to 18.7 in the baseline. Figure 5 illustrates these null results. These results confirm that simply providing a channel for intent does not automatically resolve multi-agent interference.

5 Discussion

5.1 Collision Dynamics in Shared Representation

In the Global Map Sharing condition, agents experienced 7.1 collisions per run, statistically identical to the 6.6 collisions observed in the blind Baseline B ($p = 0.83$). While shared representation granted agents perfect knowledge of the maze structure, it did not grant them knowledge of each other's future actions. Because $R_{vis} = 1$, agents separated by empty space cannot see one another. When the global map reveals a single optimal bottleneck, multiple agents optimize independently for that path and converge on it simultaneously. This indicates that static map knowledge, no matter how perfect, does not substitute for dynamic coordination mechanisms.

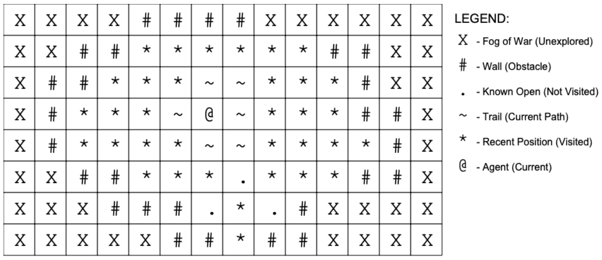

Figure 3: Partial Observability (Fog of War). Agents perceive only their immediate neighborhood (visibility radius = 1). Gray cells represent unknown territory.

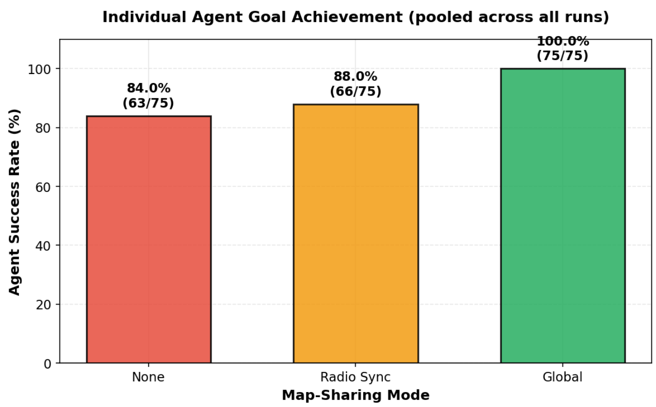

Figure 4: Impact of Map Sharing on Task Completion. Global sharing guarantees success, while Radio synchronization fails to maintain connectivity sufficient to outperform the baseline significantly.

5.2 Phenomenological Analysis of Communication

Qualitative analysis of the decision logs reveals patterns that may explain why explicit communication failed to outperform the baseline, though further experimentation is needed to isolate causal factors.

In Structured communication runs (e.g., Seed 15), agents frequently entered deadlock loops. Agent A would signal a yield; Agent B would receive this signal one turn later and signal a reciprocal yield; Agent A would receive the reciprocal yield and wait again. This stop-and-chat behavior consumed turns that could have been used for movement, potentially neutralizing the efficiency gains of avoiding collisions.

Freeform communication (Seed 13) showed more promise in terms of content—agents sent complex instructions like "please hold at (9,1)." However, these interactions appeared to suffer similar operational drag. Communicating agents moved slower than agents who simply pushed through collisions, suggesting that the overhead of coordination may exceed its benefits in this setting.

6 Conclusion

Our results delineate the distinct roles of shared state and explicit messaging in multi-agent systems. Shared representations (Maps) achieve perfect search performance, eliminating the 16% of failures observed in the blind baseline. However, they do not solve the Coordination problem; agents with perfect maps collide just as frequently as blind agents because they optimize independently for the same paths.

Explicit communication, while theoretically capable of resolving this contention, did not yield statistically significant performance improvements in our setup. Qualitative patterns in decision logs suggest that coordination overhead—including message exchange and confirmation—may consume turns that would otherwise be used for movement, though isolating the role of message latency requires further experimentation.

Beyond these specific findings, this study demonstrates `LLMGrid`'s utility as a diagnostic instrument for agent coordination research. The environment's modularity—tunable grid size, obstacle density, agent count, visibility radius, and communication bandwidth—enables researchers to stress-test coordination protocols under precisely controlled conditions. Unlike richer but noisier simulation platforms, the grid isolates mechanical friction (latency, bandwidth limits, spatial contention), enabling causal diagnosis of protocol failures. Future work can leverage this testbed to investigate synchronous communication architectures, hierarchical coordination protocols, or scaling behavior in swarms of 10+ agents.

References

[1] ARC Prize Foundation. ARC-AGI-1: The abstraction and reasoning corpus benchmark. https://arcprize.org, 2025.

[2] François Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

[3] Shuofei Hong, Tao Zhang, Zeyu Ding, Linyuan Yang, Qiang Chen, and Bo Zhang. Metagpt: Meta programming for multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*, 2023.

[4] Haoran Li, Hamed Nourkhiz Mahjoub, Behzad Chalaki, et al. Language grounded multi-agent reinforcement learning with human-interpretable communication. *arXiv preprint arXiv:2409.17348*, 2024.

[5] Yuyang Liu, Yuyang Huang, Jiayi Yao, et al. Droidspeak: KV cache sharing for cross-LLM communication and multi-LLM serving. *arXiv preprint arXiv:2411.02820*, 2025.

[6] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S.

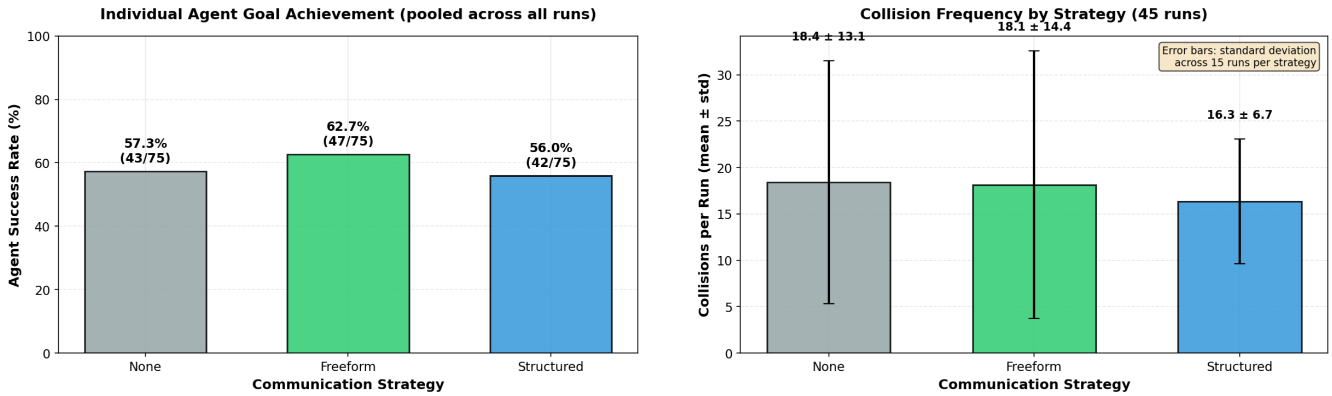

Figure 5: Inefficacy of Explicit Communication. Messaging strategies did not significantly improve success rates (Left) and failed to reduce collision rates compared to the silent baseline (Right).

Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

[7] Kyle Robison. Claude plays pokémon and the future of AI agents. *Wired*, 2025. March 12, 2025.

[8] Qiang Wang, Tianyu Wang, Qi Li, Junbin Liang, and Bingsheng He. Megaagent: A practical framework for autonomous cooperation in large-scale LLM agent systems. *arXiv preprint arXiv:2408.09955*, 2024.

[9] Zheng Wang, Yi Ren, Peng Guo, Hang Qi, and Chujie Zhang. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.

[10] Qifan Wu, Gagan Bansal, Jianfeng Zhang, et al. Autogen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.